

CONGENITAL HEART DISEASE (CHD)

Dr. Gemechu E (MD, Assistant professor of Pediatrics and Child Health)

November 19, 2023 G C.

Outlines

- To revise the fetal circulation briefly
- To discuss about the classification of congenital heart disease
- Discuss the common acyanotic and cyanotic congenital heart diseases epidemiology.
- Pathophysiology and clinical presentation
- Investigation and principles of management of common CHD

➤ **At the end of the lecture a student should be able to discuss**

- Basic fetal circulation
- Classification of CHD
- The epidemiology
- Pathophysiology
- Clinical presentation and physical findings of common CHD
- The principles of management including relevant investigations for respective lesions .

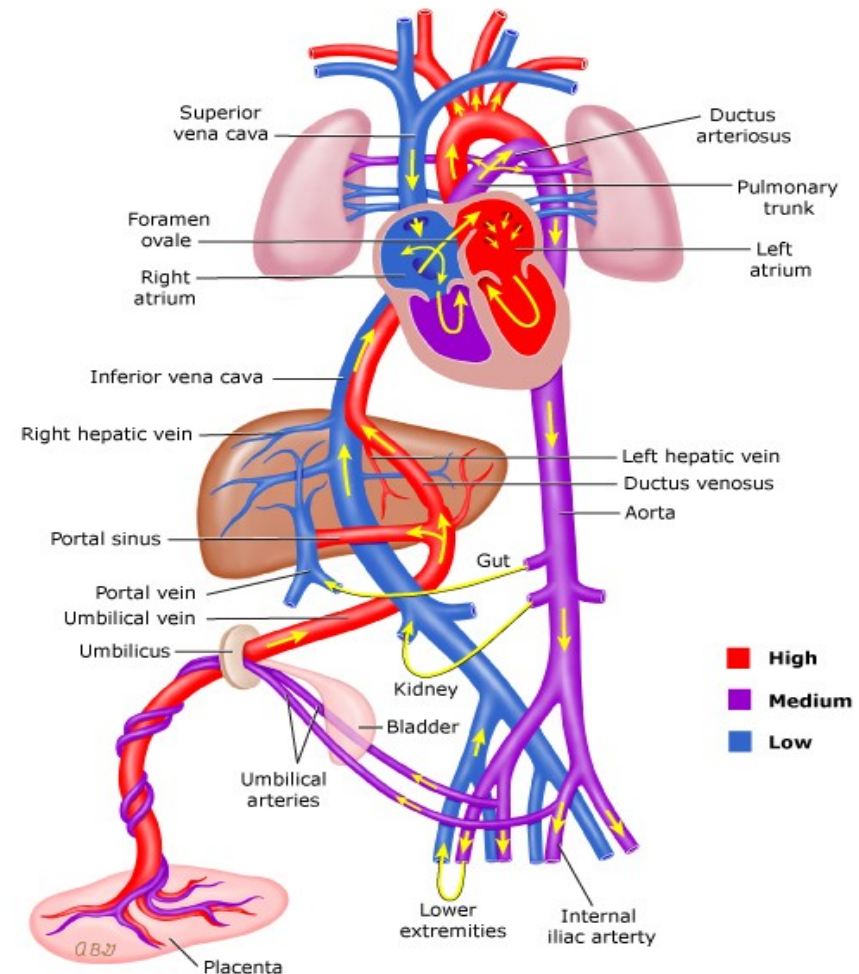
Fetal Circulation

- Fetus received Oxygen and nutrients from the mother through the placenta and umbilical cord.
- The fetus also rely on the mother to dispose of waste
- In the fetus right and left ventricles exist in a **parallel** circuit.
- Circulation relay on shunt through
 - ✓ Foramen ovale
 - ✓ Ductus arteriosus
 - ✓ Ductus venosus
- ❖ Thus it can be defined as a '**shunt-dependent**' circulation

Fetal Circulation

- ✓ Arterial blood leaves the placenta via the **umbilical vein**
- ✓ This branches and delivers blood to the IVC by way of the **ductus venosus**
- ✓ Blood then goes into the right atrium, goes across the foramen ovale
- ✓ Blood from SVC → RV via TV → RV then to PA
- Instead of going to the lungs, >85% goes through the PDA to the aorta

Fetal circulation



The degree of oxygen saturation is indicated by shading, as explained in the figure key.

Transitional circulation

Interruption of Umbilical cord:-

- Removal of the low resistance placental circulation result in an $\uparrow$ ***SVR***
- Lack of blood flow through the placenta leads to closure of **Ductus venosus** (*the ligamentum venosum*)

Expansion of lungs :

- Mechanical expansion of the lungs and increased arterial PO₂ result in a rapid **decrease in pulmonary vascular resistance.**
- The increased blood volume from the pulmonary circulation increased the LA volume and pressure sufficiently to **close the foramen ovale** (*fossa ovalis*)

PDA closure : The ductus flow become left to right and later the ductus will obliterated (*ligamentum arteriosus*)

Transitional circulation

- ✓ The right ventricle is coupled with low resistance pulmonary circulation and its wall thickness and mass decreased
- ✓ The left ventricle coupled to high resistance systemic circulation and its wall thickness and mass ↑ and deliver the entire systemic CO.
- ✓ There is a change from fetal haemoglobin to adult haemoglobin

Congenital Heart disease

• Introduction

- ✓ Structural or functional heart and major vessels disease that present at birth.
- ✓ It is not static ,there is always a continuous anatomical or physiological change
- ✓ At least 18 distinct types of CHDs are recognized, with many additional anatomic variations.
- ✓ The incidence is higher in abortus and still births
- ✓ Prevalence at birth ranges from 6 to 13 per 1000 live births

Introduction

- ✓ CHD is the most common congenital disorder in newborns.
- ❖ **The etiology - > multi factorial (genetic & environmental)**
- ✓ In less than 10% of cases the cause-effect established.
- 📁 Two to 4% associated with known environmental or adverse maternal conditions and teratogenic influences
- 📁 8% genetic
- ✓ **Critical CHD** accounts for approximately 25 % of all CHD.
- ✓ Mortality of CHD is greater during early infancy.

Introduction...

❖ Risk factors

■ Environmental factor

- ✓ Increased prevalence of PDA & ASD in children born at high altitude.
- ✓ High incidence of CHD in children of mothers who are diabetic or alcoholic.
- ✓ Congenital infection: Rubella associated with increased number of PDA and pulmonary stenosis.

Introduction....

- ✓ Prematurity: 2-3 fold higher in preterm compared with term infants excluding PDA
- ✓ Maternal factors: DM, HTN, obesity, smoking, alcohol, systemic connective tissue disorders, phenylketonuria, epilepsy...
- **Metabolic factors, genetic and chromosomal aberrations are also known to predispose to CHD.**
- ✓ Chromosomal defects were detected in 7 percent of patients with CHD, and extra cardiac anomalies in 22 percent.
- ✓ Family history – **first-degree relatives** who have nonsyndromic isolated CHD is estimated to be 3 to 4 fold higher than that of the general population
- ✓ Congenital heart defects can arise at any stage of development in intra uterine life,
- ✓ **ART:**

Common critical congenital heart defects and their association with cyanosis and dependence upon the ductus arteriosus

	Cyanosis?	Ductal-dependent?
Left-sided obstructive lesions		
Hypoplastic left heart syndrome	Yes	Yes
Valvar AS		
▪ Critical AS	Cyanosis or differential cyanosis ^{***}	Yes
▪ Moderate to severe AS	No	No
COA		
▪ Critical COA	Differential cyanosis ^{***}	Yes
▪ Moderate to severe COA	No	No
Interrupted aortic arch	Differential cyanosis* (pattern of cyanosis varies based upon type)	Yes
Right-sided obstructive lesions		
Tetralogy of Fallot	Variable	Possibly ^{††}
Tetralogy of Fallot with pulmonary atresia	Yes	Yes (unless multiple or large aortopulmonary collaterals are present)
Pulmonary atresia with intact interventricular septum	Yes	Yes
PS		
▪ Critical PS	Yes	Yes
▪ Severe PS	No	No
Tricuspid atresia	Yes	Possibly ^{††}
Severe neonatal Ebstein anomaly	Yes	Possibly ^Δ
Parallel circulations		
Transposition of the great arteries	Yes [◇]	Yes
Other		
TAPVC	Yes	No [§]
Large VSD	No	No
AV canal defect	No	No
Truncus arteriosus	Yes	No

This table summarizes the key features of some of the more common critical CHD lesions. Critical CHD refers to lesions requiring surgery or catheter-based intervention in the first year of life. The most common lesions are listed in this table; however, there are other less common congenital heart lesions that may require intervention within the first year of life.

AS: aortic stenosis; COA: coarctation of the aorta; PS: pulmonic stenosis; TAPVC: total anomalous pulmonary venous connection; VSD: ventricular septal defect; AV: atrioventricular; CHD: congenital heart disease; RV: right ventricle; PDA: patent ductus arteriosus.

* In these lesions, the upper one-half of the body (preductal) is pink and the lower one-half (postductal) is cyanotic.

†† Infants with tetralogy of Fallot or tricuspid atresia may have ductal-dependent circulation if there is severe RV outflow tract obstruction (ie, critical pulmonary stenosis or atresia).

Δ In cases of severe Ebstein anomaly with extreme cyanosis, a PDA may be necessary to maintain pulmonary blood flow until pulmonary vascular resistance drops.

◇ Reversed differential cyanosis (ie, oxygen saturation higher in the lower than upper extremity) may occur if there is coexisting coarctation of the aorta or pulmonary artery hypertension.

§ Some patients with obstructed TAPVC may require a PDA to maintain systemic cardiac output. However, a PDA may also increase the degree of cyanosis.

DIAGNOSTIC EVALUATION

❖ A cardiac examination allows

- ✓ Guide the echocardiographic evaluation toward confirming or eliminating specific diagnosis, thereby increasing its accuracy
- ✓ Eliminate unnecessary and expensive laboratory tests (innocent murmurs)
- ✓ Reassuring the patient's family and preventing unnecessary restrictions on healthy physical activity

History :

- ✓ **Antenatal history:** exposure to any teratogen, phenytoin, thalidomide, lithium, maternal DM & etc.
- ✓ **Family history:** risk of having CHD 2-6%
 - the risk is 14% if the mother has CHD.
- ✓ In most of the cases the cause is multi-factorial
- ✓ Familial occurrence of ASD, PDA, VSD, TOF & AS.

History...

- ✓ **Perinatal period**, including the presence of cyanosis, shock, respiratory distress, or prematurity.
- ✓ **Symptoms of CHF in infants and children are age specific**
- Exercise intolerance, sweating during crying or feeding, frequent interruption of feeding, shortness of breathing, poor growth and syncope.
- More severe manifestations- orthopnea & nocturnal dyspnea

Physical examination

✓ Systemic examination includes

➤ G/A, V/S, and anthropometric measurement.

- **Persistent tachycardia** (>200 beats/min in neonates, 150 beats/min in infants, or 120 beats/min in older children), bradycardia, requires investigation to exclude pathologic arrhythmias
- **A wide pulse pressure** with bounding pulses may suggest: PDA, AR, an arterial-venous communication, or increased CO
2° to anemia, anxiety, ...

Physical examination...

- ✓ **Diminished pulses:-** pericardial tamponade, left ventricular outflow obstruction, or cardiomyopathy.
- ✓ **Radial-femoral delay** → CoA
- ✓ **Blood pressure:** Difference (arms vs legs) → CoA
- ✓ **JVP**-provides information about *central venous and right atrial pressure*

Physical examination...

- ✓ **Hyperdynamic precordium** : large L-to-R shunt, thin people
- ✓ **A silent precordium**: with a barely detectable apical impulse suggests pericardial effusion or severe cardiomyopathy
- ✓ Greater than 30% of children may have an **innocent murmur** at one time in their lives
- ✓ **Absence of a murmur** does **not rule** out significant CHD

Physical examination

✓ **Cyanosis**

- ✓ Cardiac failure and chronic cyanosis result in failure to thrive
- ✓ Cyanosis is observed: nail beds, lips, tongue, and mucous membranes
- ✓ Pulse oximetry :- Pre- and post ductal pulse oximetry to assess for cyanosis and differential cyanosis.
- ✓ **Hyperoxia test** to rule out non cardiac origin of cyanosis.
- Pulmonary condition & PPH respond for 100% oxygen & $PO_2 > 150\text{mmHg}$.

Summary

➤ **CHD suspected in any child with:**

- Feeding difficulty
- Recurrent attack of respiratory tract infection
- Growth failure
- Positive pulse oximetry screening
- Cyanosis unresponsive to 100% oxygen
- Tachycardia
- Respiratory distress
- Rhythm disturbance
- Murmur (absence of murmur doesn't rule out or in CHD)

Radiology evaluation

✓ ECG

✓ CXR

✓ Echocardiography

✓ Angiography

✓ Cardiac catheterization

✓ MRI.

Echocardiography indications

- ✓ Signs or symptoms concerning for critical CHD,
- ✓ Including shock unresponsive to volume resuscitation,
- ✓ Cyanosis or differential cyanosis, failed hyperoxia test,
- ✓ Unexplained respiratory symptoms, or pulmonary
- ✓ ECG and/or chest radiograph findings suggestive of CHD
- ✓ P/E findings suggestive of CHD, including abnormal heart sounds (eg, S3 gallop, single S2, click), pathologic murmur, diminished or absent lower extremity pulses, abnormal 4 extremity BP.
- ✓ Positive pulse oximetry screening
- ✓ Genetic disorder or extra cardiac malformation associated with cardiovascular malformations

CLASSIFICATIONS

✓ Evaluation of the infant or child with CHD has 3 major components:

❖ 2 major groups **cyanotic** and **acyanotic**

- These 2 groups further sub-divided with CXR evidence of ↑sed, normal or ↓sed pulmonary vascular markings.

❖ For diagnostic purpose can be classified as :

- Left to right shunt
- Right to left shunt
- Obstructive lesions

1. Acyanotic CHD with left to right shunt

*** Shunting of fully oxygenated blood back in to the lung.**

- ✓ Ventricular septal defect(VSD) 25-30%
- ✓ Atrial septal defect (6-8%)
- ✓ Patent ductus arteriosus (6-8%)
- ✓ Atrio ventricular septal (AV canal) defect

2. Acyanotic CHD with obstructive lesions

- ✓ Pulmonary stenosis (7-10%)
- ✓ Coarctation of aorta (5-7%)
- ✓ Congenital Aortic stenosis (4-7%)
- ✓ Aortic arch anomaly
- ✓ Hypoplastic left heart syndrome.
- ✓ Ebstein anomaly

3. CYANOTIC CHDs:

1. Tetralogy of Fallot/TOF (5-7%)
2. Transposition of great arteries(TGA)
3. Truncus arteriosus
4. Tricuspid atresia
5. Total anomalous pulmonary venous connection.

Case 1

Fatuma is 3 month old infant presented at OPD with a c/c of fast breathing and excessive sweating specially when she feeds on breast . She has no fever , she did not gain adequate weight for her age . She was treated for pneumonia 2 times since the last 2 months .

P/E : She is tachypneic, tachycardic, clear chest. she has active precordium, shifted PMI, no audible murmur, 4cm palpable liver BRCM.

1. Acyanotic CHD with left to right shunt

1.1. VSD

- ✓ The most common form of CHD (25% of all CHD).
- ✓ Male > female
- ✓ Can be at membranous (commonest type) or muscular type
- ✓ The size of the VSD is major determinant of the left-right shunt.
- ✓ QP: QS is 1:1 the flow is bidirectional, when it reach QP: QS is 2:1 left atria & ventricle volume over load.

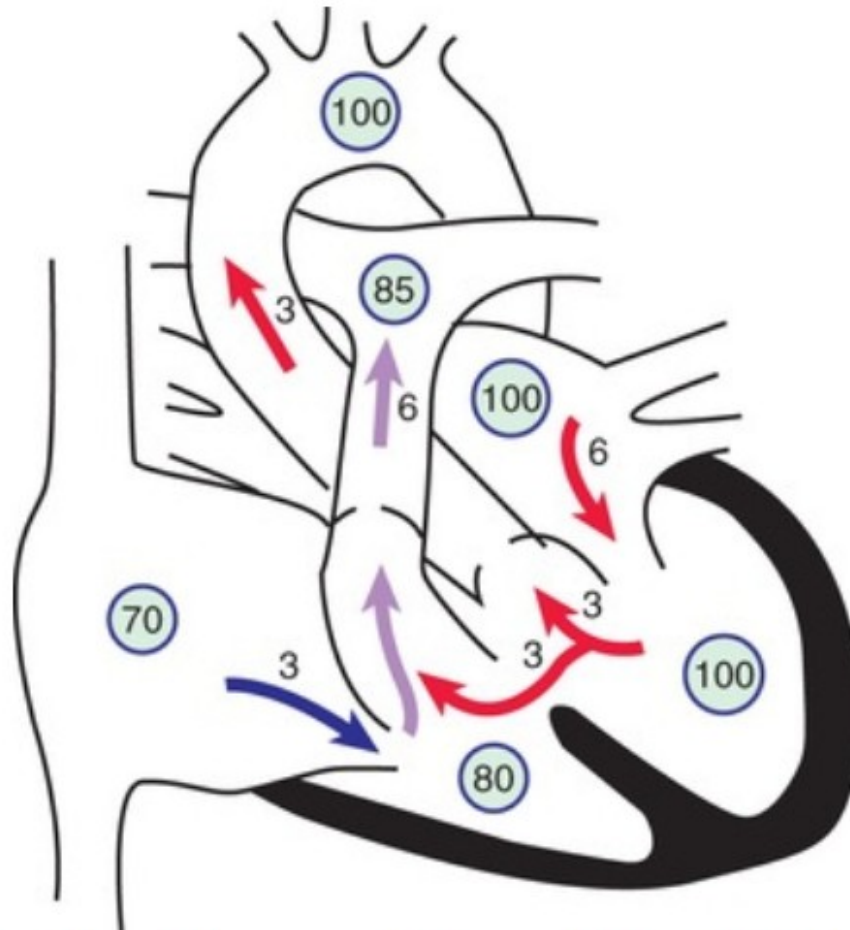


FIG. 453.8 Physiology of a large ventricular septal defect. *Circled numbers* represent oxygen saturation (SO_2) values. The *numbers next to the arrows* represent volumes of blood flow (in $L/min/m^2$). This illustration shows a hypothetical patient with a pulmonary-to-systemic blood flow ratio ($Q_p:Q_s$) of 2 : 1. Desaturated blood enters the right atrium from the vena cava at a volume of 3 $L/min/m^2$ and flows across the tricuspid valve. An additional 3 L of blood shunts left to right across the VSD, resulting in increased SO_2 in the right ventricle. Six liters of blood is ejected into the lungs. Pulmonary arterial saturation may be further increased because of incomplete mixing at right ventricular level. Six liters returns to the left atrium, crosses the mitral valve, and causes a mid-diastolic flow rumble. Three liters of this volume shunts left to right across the VSD, and 3 L is ejected into the ascending aorta (normal cardiac output).

VSD cont....

Types of VSD

- 70% perimembranouse
- 5-20% Muscular
- 5-8% sub pulmonic (Subarterial)(more in south east Asia
- 8-10% AV canal septum type (inlet)

Clinical presentation

❖ Severity depend on :

- **Size of the defect**
- **Level of pulmonary to systemic resistance.**
- Majority of Small restrictive lésions closes spontaneously
- Large defects lead to CHF at early age

VSD- Large

- ✓ Lesion close to aortic root size $\geq 6\text{mm}$
- ✓ Dyspnea, feeding difficulty, poor growth, recurrent LRTI, & cardiac failure in early infancy.
 - ✓ Parasternal heave, less intense shorter murmur.
 - ✓ Pulmonary and right ventricular hypertension.
 - ✓ Mid diastolic rumbling murmur at the apex caused by increased blood flow at the mitral valve' which indicate the ratio of QP: QS is $>2:1$.
- ✓ **CXR:** gross cardiomegaly. Prominence of ventricle, left atria, and pulmonary artery, sign of pulmonary edema with effusion.

Chest X-Ray (VSD)

- Cardiomegaly
(LA, LV, RV, PA)
- Increased vascular
marking

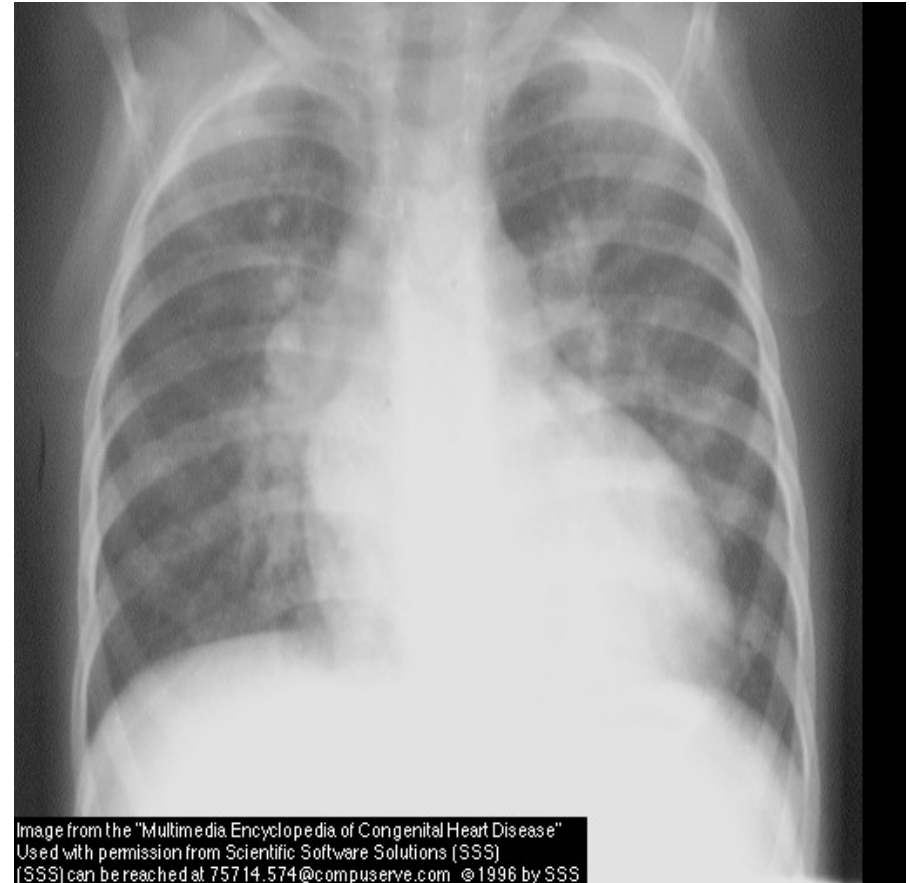
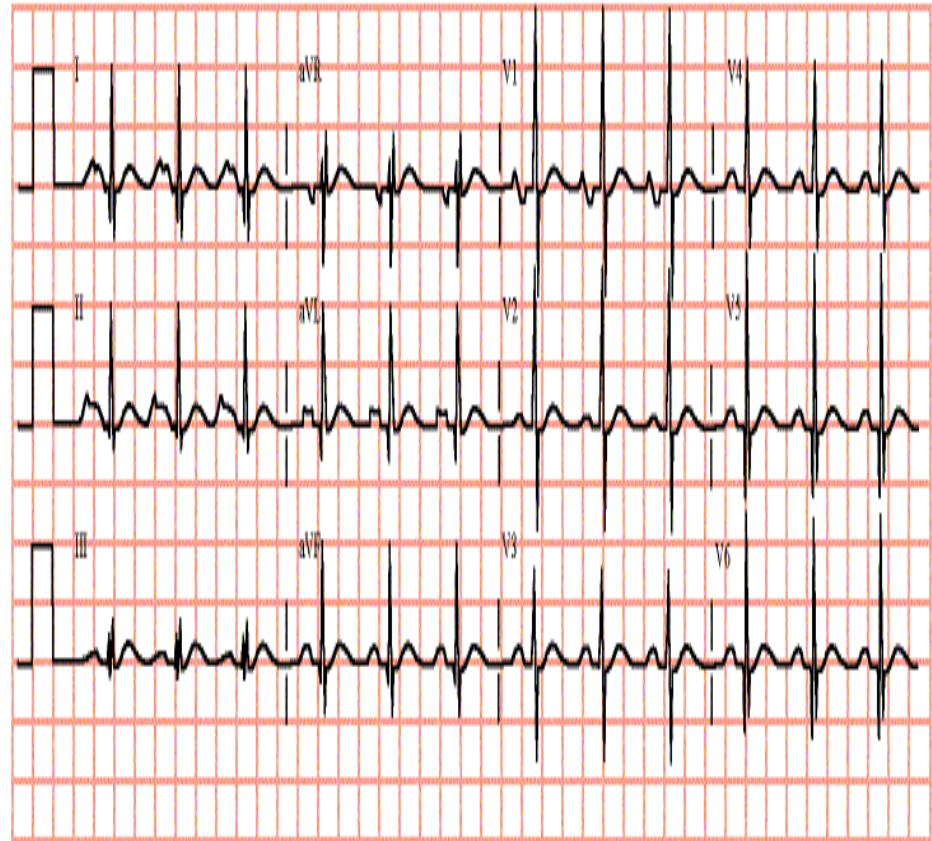


Image from the "Multimedia Encyclopedia of Congenital Heart Disease"
Used with permission from Scientific Software Solutions (SSS)
(SSS) can be reached at 75714.574@compuserve.com ©1996 by SSS

Diagnostics cont...

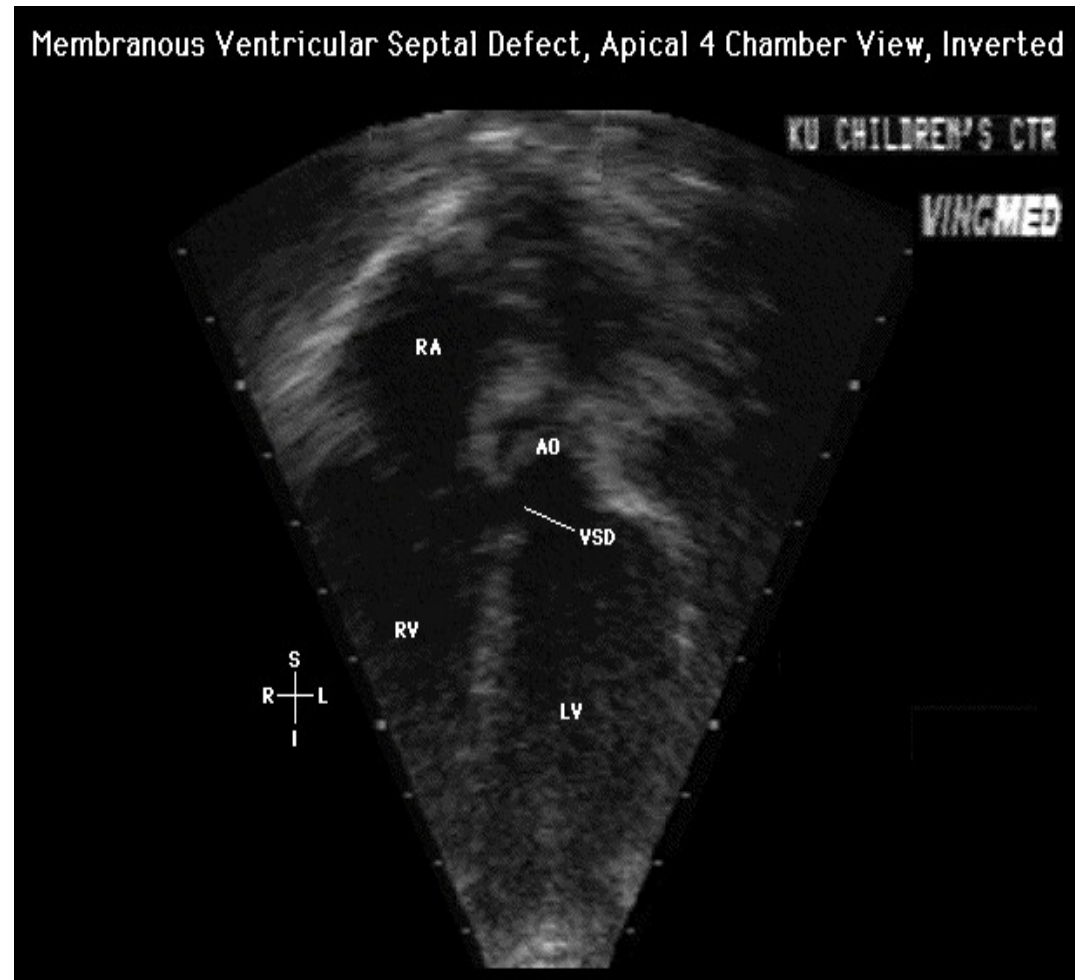
ECG:

- P wave peaked and notched
- Left ventricular hypertrophy or biventricular



Echocardiography (VSD)

- Position and size of VSD
- Chambers size
- Pressure gradient across the defect
- Direction of shunt



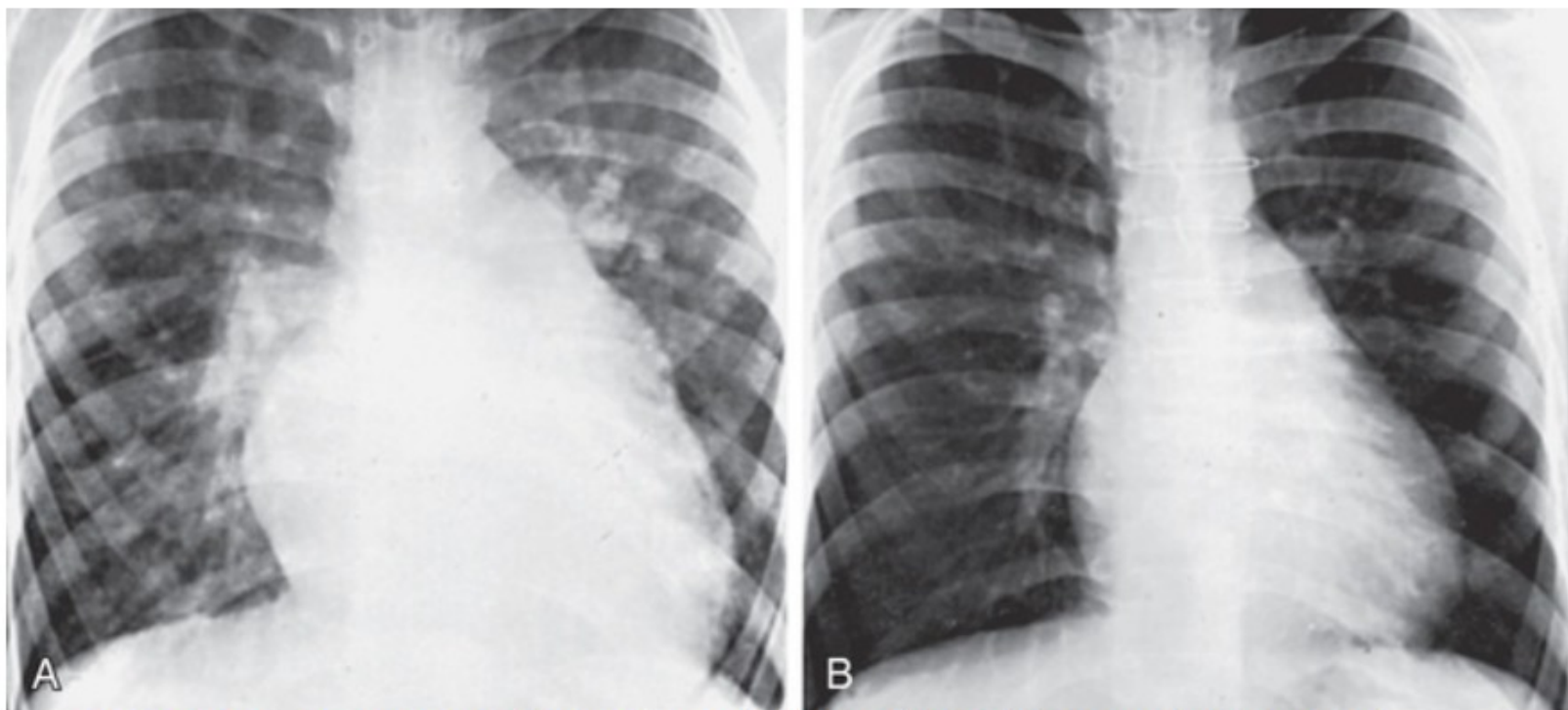


FIG. 453.9 **A**, Preoperative radiograph in a patient with a ventricular septal defect with a large left-to-right shunt and pulmonary hypertension. Significant cardiomegaly, prominence of the pulmonary arterial trunk, and pulmonary overcirculation are evident. **B**, Three years after surgical closure of the defect, heart size is greatly decreased, and the pulmonary vasculature is normal.

VSD...

Prognosis and complications:

- ✓ 8% of large VSD closed spontaneously.
- ✓ Repeated LRTI, Endocarditis, cardiac failure & Eisenmenger physiology are common complications.
- ✓ Small VSD in adults show increased episodes of arrhythmia, sub aortic stenosis and exercise intolerance.

1.1. VSD...

TREATMENT: Small VSD follow up and management:-

- ✓ The child encouraged to live normal life with out restriction.
- ✓ Prevention for endocarditis.
- ✓ Echocardiography to screen for spontaneous closure and for the development of left ventricular out flow tract pathology (AS, AR).
- ✓ Medical treatment has two aims
 - To control heart failure.
 - To prevent the development of pulmonary vascular disease.

1.1. VSD...

Surgery : Indications for surgery are:-

- large defects with clinical symptom and FTT can't be controlled by medical treatment.
- Infants between 6-12 month of age with large defect and associated with pulmonary hypertension.
- Patients older than 2 years with QP: QS ratio $> 2:1$.
- Supracrystal VSD of any size to avoid risk of AR.
- ✓ **Severe pulmonary hypertension** is a contraindication for surgery.
- ✓ After surgery: All symptom abolish, pulmonary hypertension regress, cardiac medications no longer required, catch up growth in 1-2years

1.2. ASD:

- ✓ An opening in atrial septum permitting the shunting of blood between the two atria.
- ✓ Three major types;
 - SINUS VENOSUS Type: high in the atrial septum. Least common form.
 - OSTEUM SECUNDUM Type; the most common type. (10% of CHD)
 - OSTEUM PRIMUM Type: low in position at AV junction.
 - Associated with incompetence of anterior leaf let of MV so prevention for endocarditis is necessary.

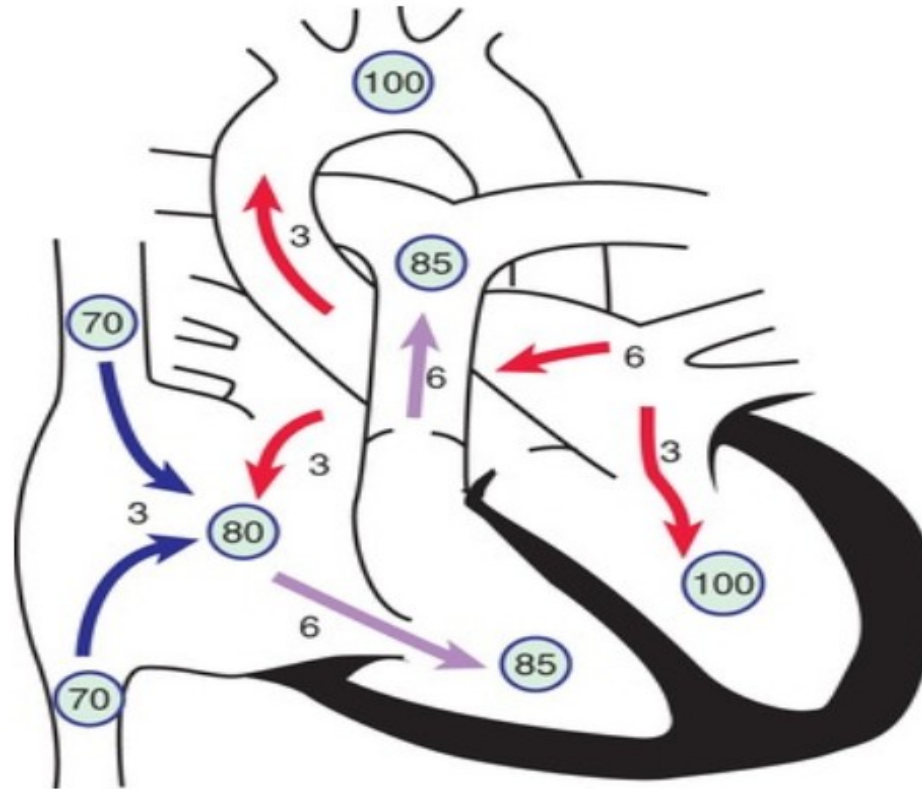


FIG. 453.2 Physiology of atrial septal defect (ASD). *Circled numbers* represent oxygen saturation (SO_2) values. The *numbers next to the arrows* represent volumes of blood flow (in $L/min/m^2$). This illustration shows a hypothetical patient with a pulmonary-to-systemic blood flow ratio ($Q_p:Q_s$) of 2 : 1. Desaturated blood enters the right atrium from the vena cava at a volume of 3 $L/min/m^2$ and mixes with an additional 3 L of fully saturated blood shunting left to right across the ASD; the result is an increase in SO_2 in the right atrium. Six liters of blood flows through the tricuspid valve and causes a mid-diastolic flow rumble. SO_2 may be slightly higher in the right ventricle because of incomplete mixing at the atrial level. The full 6 L flows across the right ventricular outflow tract and causes a systolic ejection flow murmur. Six liters returns to the left atrium, with 3 L shunting left to right across the defect and 3 L crossing the mitral valve to be ejected by the left ventricle into the ascending aorta (normal cardiac output).

1.2. ASD

- ✓ Pulmonary to systemic blood flow is between 2:1 to 4:1.
- ✓ Patients may have dyspnea on exertion
- ✓ Pulmonary vascular resistance remains low throughout childhood.
- ✓ No murmur across ASD.
- ✓ Diastolic rumbling murmur at the tricuspid valve with systolic ejection murmur & S2 split.

1.2. ASD:

- ✓ CXR: Increase pulmonary artery vascularity.
- ✓ ECG & ECHO: reveal pure right ventricular enlargement.
- ✓ Prognosis and complication: secundum ASD is well tolerated up to the 3RD decade, endocarditis is rare & prophylaxis is not recommended.
- ✓ Treatment: Surgery or Tran catheter device closure.

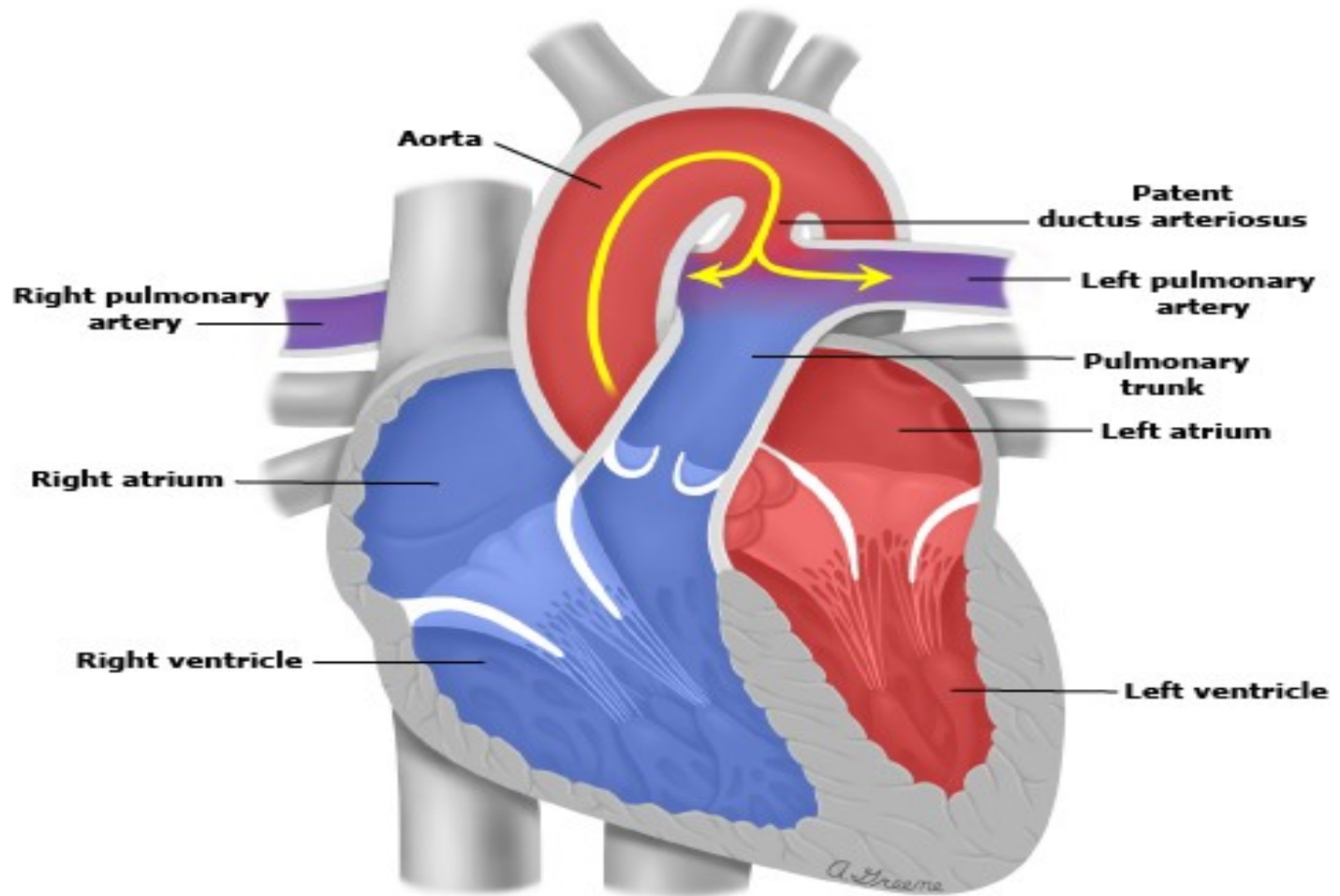
Case 2

- A 3 months old infant presented at the OPD with a c/c of fast breathing and excessive sweating specially when she is feeding . She has no fever , she did not gain weight. She was treated as a case of pneumonia two times since the last four weeks .
- On P/E has no cyanosis. She is tachycardic and in respiratory distress. Clear chest. CVS pericordial : Active precordium, shifted PMI, continuous murmur at the right upper sternal boarder . palpable liver 3 cm BRCM.

1.3. PATENT DUCTUS ARTERIOSUS(PDA).

- is an abnormal persistence of a normal fetal pathway between the pulmonary artery and the aorta.
- **usually closes during the first 24-72hours**
- is seen in 10% of other CHD,
- Females :males 2:1.

Patent ductus arteriosus



Graphic 65828 Version 2.0

PDA...

- ✓ Common in premature infants and following congenital infection (rubella virus).
- ✓ In term infants the PDA walls deficient in both mucoid, endothelial and the muscular media.
- ✓ But in premature infant the PDA wall has normal structure the patency is due to hypoxia and immaturity

CLINICAL MANIFESTATIONS

- Determined by the degree of left-to-right shunting,
 - ✓ which is dependent upon the size and length of the PDA, and the difference between pulmonary and systemic vascular resistances.
- The hemodynamic consequences of the PDAs can be categorized by the degree of left-to-right shunting based upon the Qp:Qs ratio.
 - ✓ Small – Qp:Qs <1.5 to 1
 - ✓ Moderate – Qp:Qs between 1.5 and 2.2 to 1
 - ✓ Large – Qp:Qs >2.2 to 1

PDA...

- ✓ The major physiologic consequence of the shunt is left atrial and left ventricular volume over load.
- ✓ Large ductus:
 - Up to 70% of left ventricular out put shunted.
 - Congestive heart failure is common.
 - Has a Risk of developing pulmonary vascular resistance.
 - A **wide systemic pulse pressure** that produce bounding pulse.

✓ **Large PDA:**

- The left ventricular impulse is hyper dynamic.
- Systolic thrill and continuous machinery murmur at the left upper sternal border..
- Low pitched rumbling mid diastolic mitral valve murmur at the apex.
- Prominent right ventricular impulse in the presence of pulmonary hypertension & differential cyanosis.

PDA...

- ✓ ECG: Left or biventricular enlargement/deep Q and tall T-wave
- ✓ CXR: cardiomegaly with left ventricular silhouette.
 - Plethoric lung.
- ✓ Echocardiography: identify the PDA.
- ✓ Cardiac catheterization: To rule out aortico-pulmonary fistula

➤ Assessment of severity:-

- The larger the heart size the larger the left to right shunt.
- Absence of 3RD sound and delayed diastolic murmur indicate small shunt.
- The wider the pulse pressure the larger the shunt.

➤ Small ducts:-asymptomatic, normal pericardial activity, other findings are variable.

✓ECG and CXR's are normal.

TREATMENT ;

- Medical therapy:
 - Congestive heart failure treatment
 - infective endocarditis prophylaxis
 - Closure of the PDA coil embolization or device closure without thoracotomy
- Surgical closure of the PDA (ligation)

2. Obstructive Lesions

▪ Hemodynamic Alterations

- ✓ increasing the resistance to ejection
- ✓ a higher systolic pressure to force a normal cardiac output through the obstructive lesion.
- ✓ Pressure difference between the ventricle and the great artery and an increase in the muscle mass of the affected ventricles.

2.1. pulmonary stenosis with intact ventricular septum.

- ✓ The most common type of obstructive lesions.(7-10% of all CHDs.)
- ✓ No symptom with mild &moderately severe disease.
- ✓ Right ventricular lift.
- ✓ Grade 1/6-4/6 systolic ejection murmur at pulmonary area
- ✓ S2 widely split with soft to inaudible P2.
- ✓ Cyanosis and high incidence of right side heart failure in severe cases.

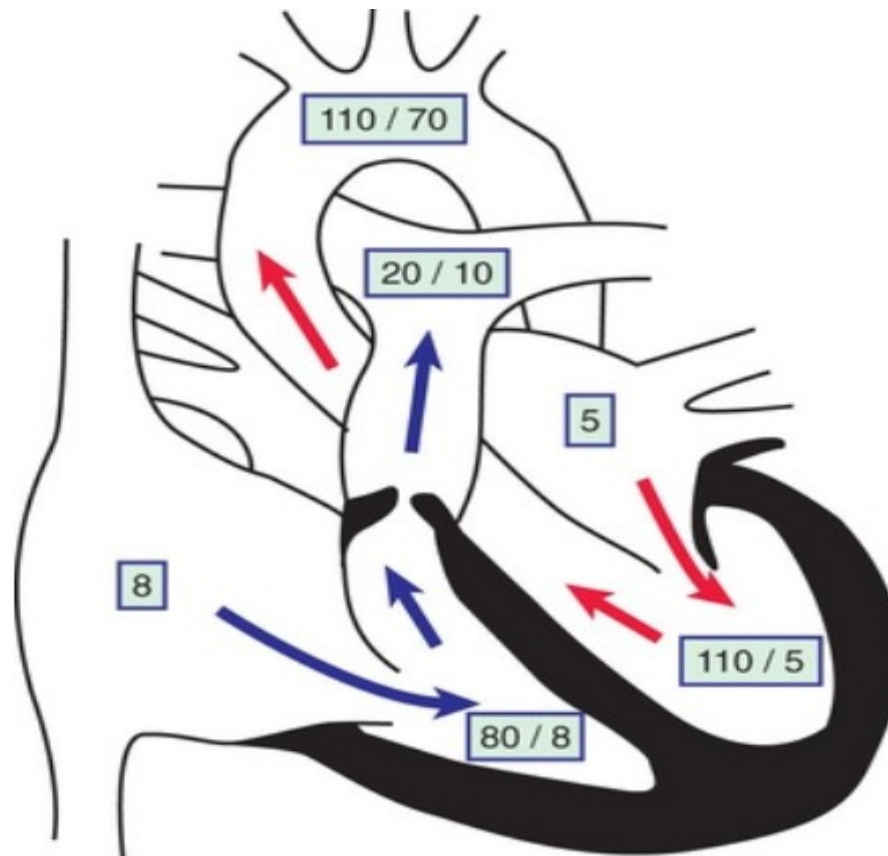


FIG. 454.1 Physiology of valvular pulmonary stenosis. *Boxed numbers* represent pressure in mm Hg. Because of the absence of right-to-left or left-to-right shunting, blood flow through all cardiac chambers is normal at 3 L/min/m². The pulmonary-to-systemic blood flow ratio (Qp:Qs) is 1 : 1. Right atrial pressure is increased slightly as a result of decreased right ventricular compliance. The right ventricle is hypertrophied, and systolic and diastolic pressure is increased. The pressure gradient across the thickened pulmonary valve is 60 mm Hg. The main pulmonary artery pressure is slightly low, and poststenotic dilation is present. Left-sided heart pressure is normal. Unless right-to-left shunting is occurring through a foramen ovale, the patient's systemic oxygen saturation will be normal.

Pulmonary stenosis/PS

- Depending on the severity classified as :-
 - Mild PS: right atrial pressure 10- 30mmHg
 - Moderate PS: right atrial pressure gradient between 30-60mmHg.
 - Severe PS: mild to moderate cyanosis with sign of right side heart failure.

Pulmonary stenosis...

✓ Prognosis and complications

- Heart failure occurs in very severe cases usually in the first month of life.
- Endocarditis is a risk but not common in childhood.

Treatment

- Moderate to severe pulmonary stenosis needs relief of obstruction.
 - Balloon valvuloplasty
 - Surgery; for severely thickened pulmonary valves.
- ✓ *No difference in patient status after valvuloplasty or surgery.

2.2. Left ventricular out flow obstructions:

2.2.1. Congenital aortic stenosis

- ✓ Account 5% of CHD
- ✓ Can be one of the three types
 - Valvular stenosis: most common type.
 - Sub valvular stenosis: beneath the aortic valve.
 - Supra valvular stenosis: above the aortic valve.
- ✓ Sub valvular stenosis may progress rapidly in severity .usually occurs with other CHD

AS...

✓ **Physiologic importance**

- Impose resistance to left ventricular emptying.
- Increased left ventricular muscle mass
- Compliance decreases and diastolic pressure rises.
- CHF and disproportional increase in coronary circulation (blood supply & demand imbalance) -> myocardial ischemia.
- Generalized fibrosis may result from a long standing supply-demand imbalance.
- The sub endocardium is particularly vulnerable to ischemia as a result papillary muscle infarction and mitral regurge may occur.

AS...

- ✓ Mild to moderate obstruction does not lead to symptom.
- ✓ Severe obstruction during infancy also known as critical aortic stenosis and is associated with left ventricular failure or profound low cardiac output and shock.
- ✓ Older children and adolescents with **severe** AS have symptoms of angina pectoris and may experience episodes of syncope.
- ✓ Louder, harsh and longer murmur indicates greater degree of obstruction.

AS...

- ✓ ECG: LVH & inverted T-wave on left pericardial lead.
- ✓ CXR; Prominent Ascending Aorta, no cardiomegaly, valvular calcification on older children.
- ✓ Echo: ventricular hypertrophy thickened and domed aortic valve. Other anomalies in 20% of patients with AS.
- ✓ Cardiac catheterization: measure the magnitude of the pressure gradient from the left ventricle to the aorta.

AS...

- ✓ Prognosis: Progression of disease seen in 5-10yr for mild to moderate ds.
- ✓ Infants with critical aortic stenosis may have severe heart failure and deteriorate rapidly to shock state.
- ✓ Based on aortic gradient can be mild ds (b/n 40-50mmHg), moderate disease (b/n 50-70mmHg) and severe ds $>70\text{mmHg}$)
- ✓ Sudden death is a significant risk for severe AS.
- ✓ AS is a Common Cause of Sudden Death in children so avoid competitive sports in children with moderate to severe disease .

AS...

TREATMENT:

✓ Balloon valvuloplasty: for mod. to severe disease.

Indication:-

1. When systolic gradient between aorta and left ventricle exceeds 60-70mmHg.at rest.
 2. For symptomatic patient and ECG changes lesser gradient<60mmHg.
 3. For rapidly progressive disease gradient of 40-50mmHg is operable.
- ✓ Surgery; for neonatal age due to inaccessible arteries and sub AS.
- ✓ Life time prophylaxis for Endocarditis.

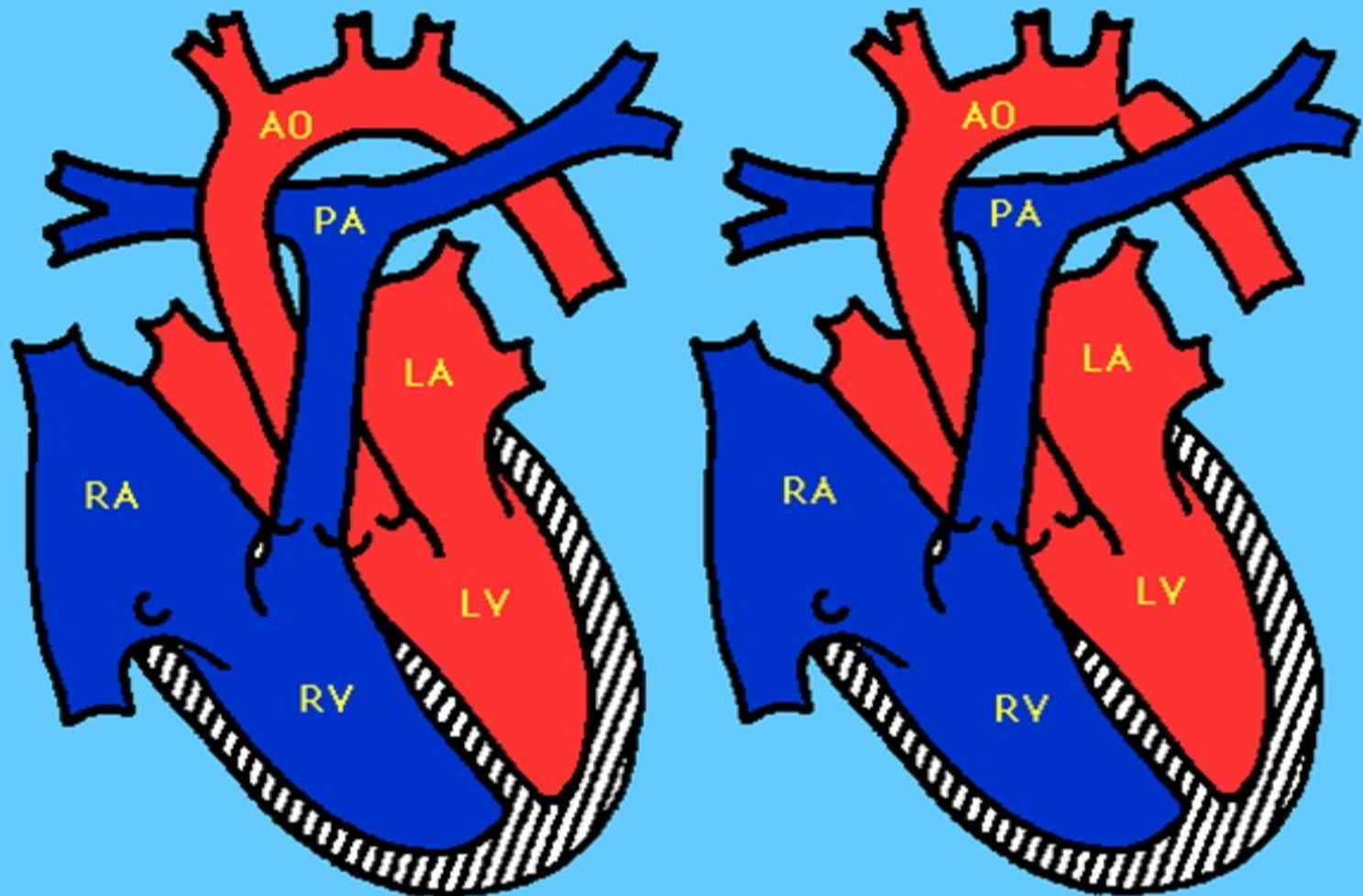
Case 3

- Chala is 8 yrs old seen at the clinic for URTI and when the nurse took the vital sign she noticed a high blood pressure and notified the physician .
- The physician examined him and he found the BP discrepancy between upper and lower extremities and checked the pulse and identified a strong pulses in the upper extremities but very weak pulse at the lower extremities . No other abnormal physical finding.

2.2.2.COARCTATION OF THE AORTA.

- ✓ A localized narrowing of the aorta either proximal (preductal) or distal (post ductal) to the ductus.
- ✓ Simple coarctation of aorta almost always (in 98% of the cases) is juxta ductal.
- ✓ account 6% of all CHD, common in males M: F 3:1.
- ✓ In preductal coarctation, the ductus remains open and shunts blood from the pulmonary artery through the ductus arteriosus into the aorta.

Coarctation of the Aorta



Normal

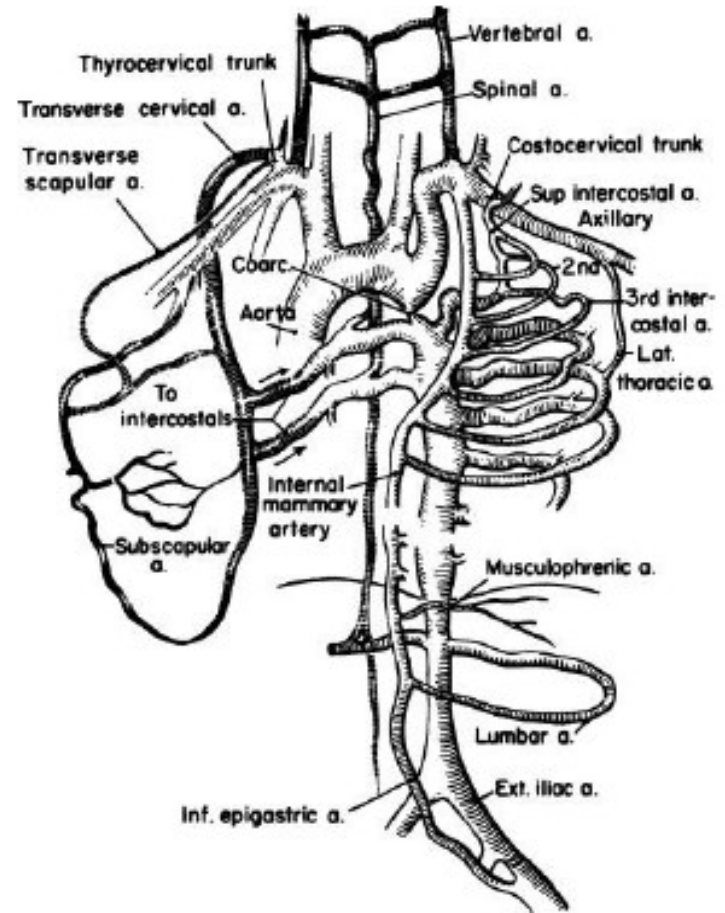
Coarctation of the Aorta

COARCTATION...

- ✓ In 70% of patients bicuspid aortic valve.
- ✓ Mitral valve abnormality, subaortic stenosis are a potential associated lesions.
 - When coarctation occurs with these lesions called *Shone complex*
- ✓ In post ductal coarctation and in mild juxta ductal type (adult type) symptoms do not arise till late adolescent or adult life.

Pathophysiology

- In case of severe COA when the ductus closed flow to the head and neck vessels is maintained.
- Flow to the lower body distal to the coarctation site is dramatically reduced and may lead to shock.
- **As the result:**
 - Elevation of systolic pressure in proximal aortic segment. (Upper extremities)
 - ❖ Vasoconstriction of arterioles to maintain diastolic pressure
 - ❖ Collateral formation to by pass the obstruction .



Clinical Manifestations

- Severe critical stenosis the neonate present with evidence of CHF if not corrected surgically result in death.
- Past the neonatal period Usually asymptomatic
- Older children:
 - ✓ Headaches
 - ✓ Epistaxis
 - ✓ Claudication, cold feet

Physical Examination:

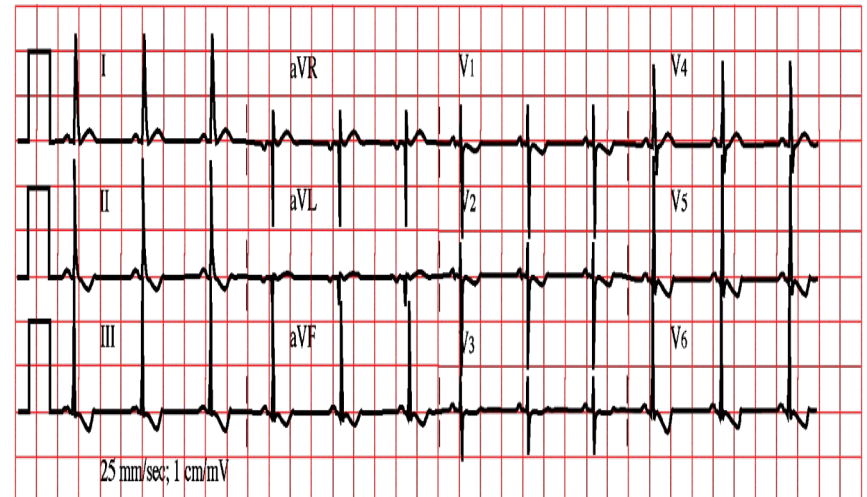
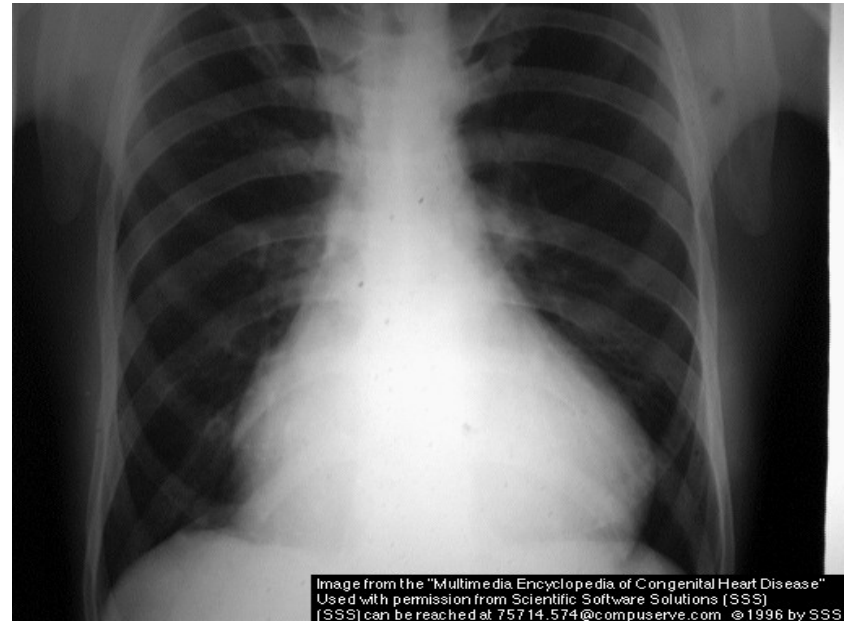
- Weak or absent femoral pulses
- **Increased B/P** in the upper extremities
- **B/P difference** between upper and lower extremities
- Radio Femoral pulse delay (Collaterals)
- A2 is loud , systolic murmur 3rd and 4th ULNB (if associated bicuspid aortic valve)

Cont....

➤ **Chest x-ray:**

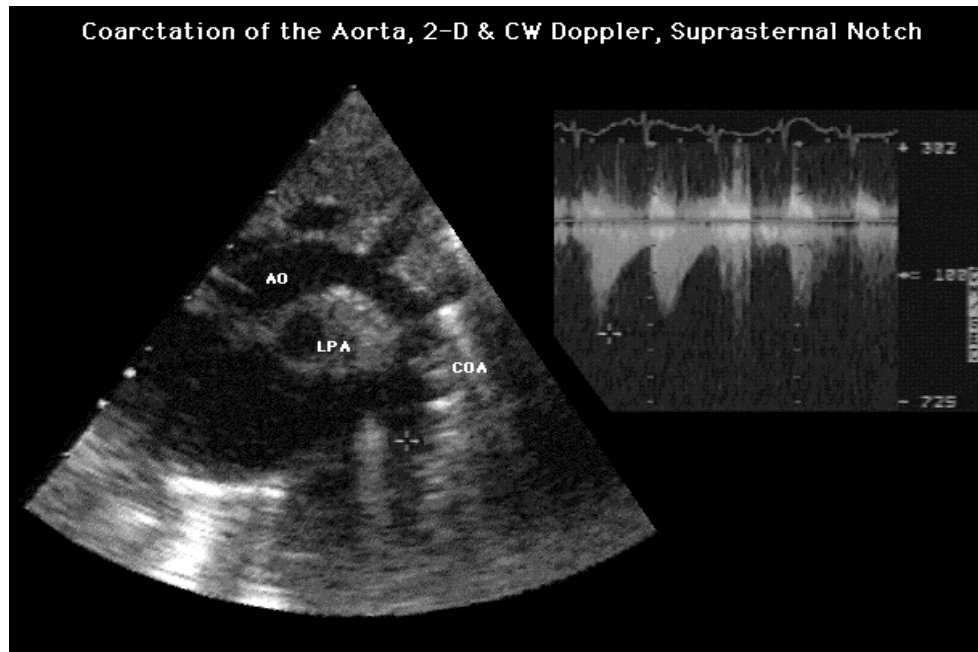
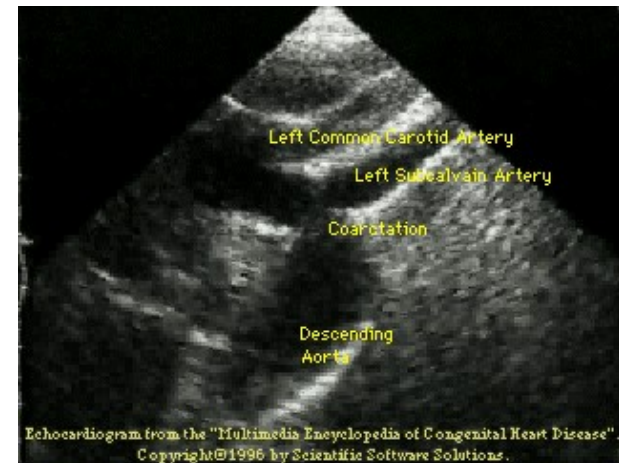
- Dilated descending Aorta, enlarged LV.
- Rib notching

➤ **ECG:** normal in childhood, later LV hypertrophy.



ECHO

- Measure the stenotic area
- gradient



TREATMENT:

- ✓ Medical treatment of hypertension before surgery otherwise risk of post operative malignant hypertension.
- ✓ In neonates with severe coarctation closure of the ductus often result hypo perfusion, acidosis and rapid deterioration.
 - PGE1 should be given .
- ✓ Surgery remain the treatment of choice
- ✓ Balloon angioplasty.

3. CYANOTIC CONGENITAL HEART DISEASE

- ✓ Right to left shunt causes desaturation of systemic venous blood to enter the systemic arterial circulation.
- ✓ Three groups of cyanotic heart lesions.
 - Admixture lesions complete mixing of the systemic and pulmonary venous circuit in atrium, ventricle or great artery (e.g. Truncus. A)
 - Pulmonary hypo perfusion:-reduced pulmonary blood flow and right to left shunt. (e.g. TOF)
 - Transposition of great arteries: the pulmonary and systemic circulations are separate and parallel.

3.1. Tetralogy of Fallot(TOF)

- ✓ The commonest cyanotic congenital heart disease.
- ✓ Has 4 essential components:
 - Ventricular septal defect(large non restrictive VSD)
 - Pulmonary stenosis.
 - Overriding of aorta over the ventricular septum.
 - Right ventricular hypertrophy.

TOF...

- ✓ In few infants the pulmonary stenosis may be less severe, associated initially with a left to right shunt→ pink TOF.
- ✓ Cyanosis usually develops few months after birth.
- ✓ Cyanotic spells/paroxysmal hyper cyanotic attacks, hypoxic blue or “Tet’ spells are problem during the first two years of life.
- ✓ The child assumes squatting position on exercise

TOF

- ✓ Exercise and crying increase venous return and decrease systemic resistance which increases the cyanosis.
- ✓ Older children have clubbing.
- ✓ Heart failure is very rare.
- ✓ No clinical cardiomegaly, but signs of right ventricular hypertrophy are present.
- ✓ Systolic thrill and ejection systolic murmur at the pulmonary area.

- ✓ECG: RVH and peaked p-wave.
- ✓CXR: Oligemic lung, Boot shaped heart, heart not enlarged, concavity is found in the area usually occupied by the pulmonary artery. Ascending aorta is large and there is right aortic arch.

Complication:

1. Cerebrovascular accident (CVA), slugging and insitu thrombosis(Cerebral thrombosis) Common in less than 2years.

Complications of TOF...

- 2. Brain abscess usually in children >2years of age less common than CVA.
- 2. Anemia of iron deficiency makes the RBC somewhat stiffer and lead to less oxygen delivery.
- 2. Bacterial Endocarditis.
- 2. Paroxysmal Hypoxic spells
- 2. Retardation of growth and Development
- 2. Hemoptysis may result from thrombosis in the pull artery and/or due to platelet dysfunction

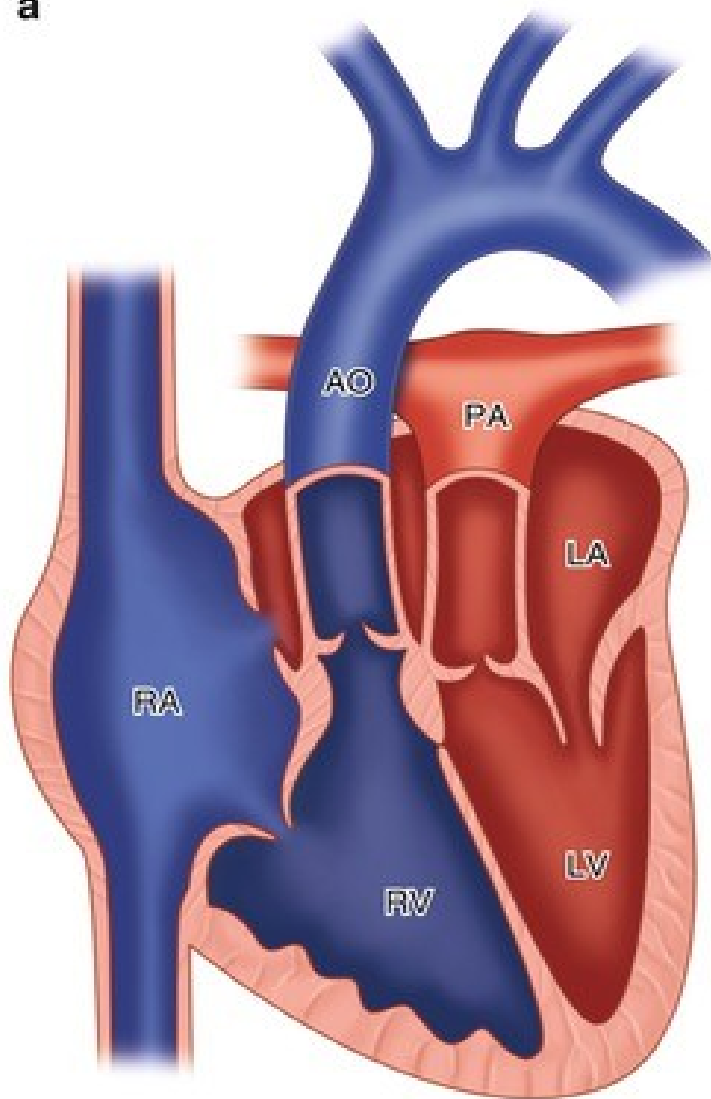
Treatment

- ✓ Medical treatment for hyper cyanotic attacks
- 1. Placement of infants in knee chest position.
- 2. 100% O₂ administration.
- 3. Morphine 0.2 mg /kg SC.
- 4. Propranolol 0.1-0.2mg/kg IV may help severe spells then 0.5mg/kg/6hr orally.
- 5. Correct acidosis and anemia.
- 6. Consider operation.
- ✓ Surgery: - complete repair is recommended before 6 months of age.

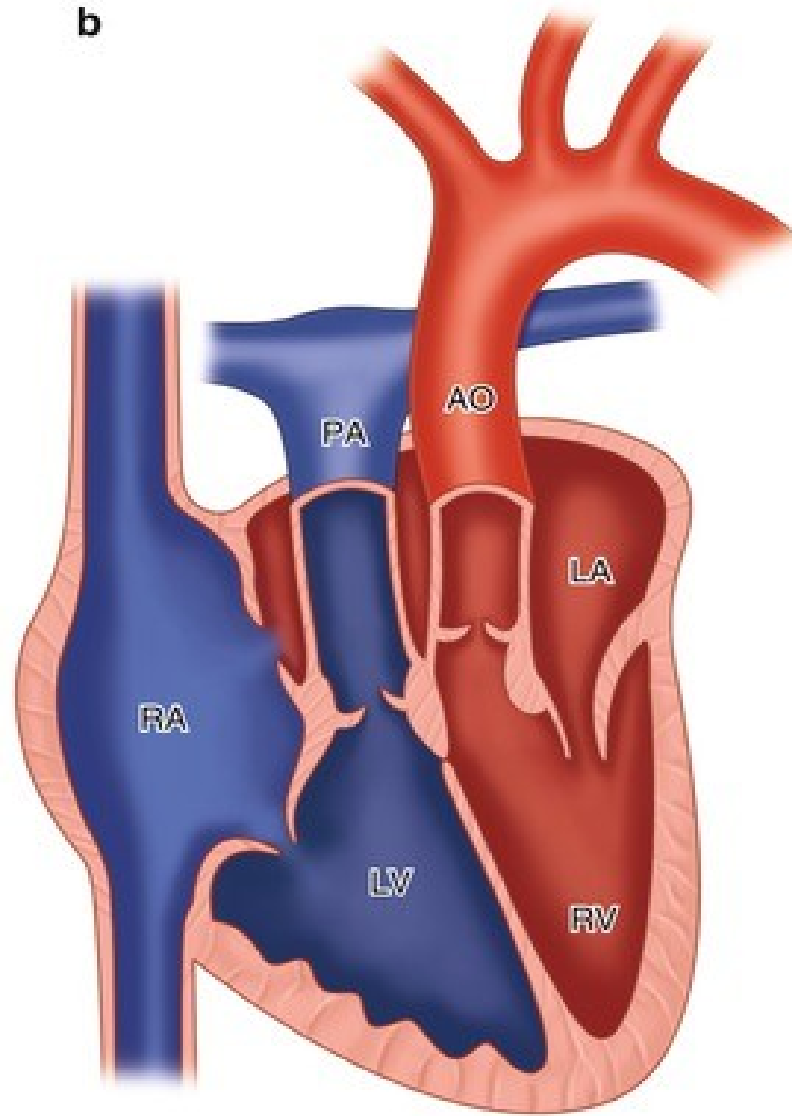
3.2. TRANSPOSITION OF GREAT ARTERIES (TGA)

- ✓ The commonest CHD Causing cyanosis shortly after birth and responsible for majority of deaths from cyanotic CHD in the 1st year of life.
- ✓ The great arteries arise from the inappropriate ventricles, the aorta from the right and the pulmonary artery from the left.
- ✓ Survival is depending on mixing of blood between the circulations through VSD, ASD, or PDA.

a



b



TGA...

- ✓ Associated abnormality of the atrial and ventricular septum, AV valves and coronary arteries are common.
- ✓ TGA with out VSD is the common variant. After birth mixing between the parallel circulations diminishes as the ductus arteriosus closes.
- ✓ Severe hypoxemia, acidosis, and death can occur rapidly.
- ✓ Cyanosis usually noted in the 1st day of life and progress rapidly as the ductus arteriosus closes.

TGA...

- ✓ Differential cyanosis may be present.
- d-Transposition is the only condition that leads to preferential desaturation of the upper body.
- ✓ S2 is loud because the transposed aorta lies anteriorly close to the chest wall.
- ✓ No murmur usually or grade 2 systolic ejection murmur.
- ✓ ECG: RVH and right axis deviation.
- ✓ CXR: Cardiomegaly with a narrow base and plethoric lung fields. Gives egg on stick picture.



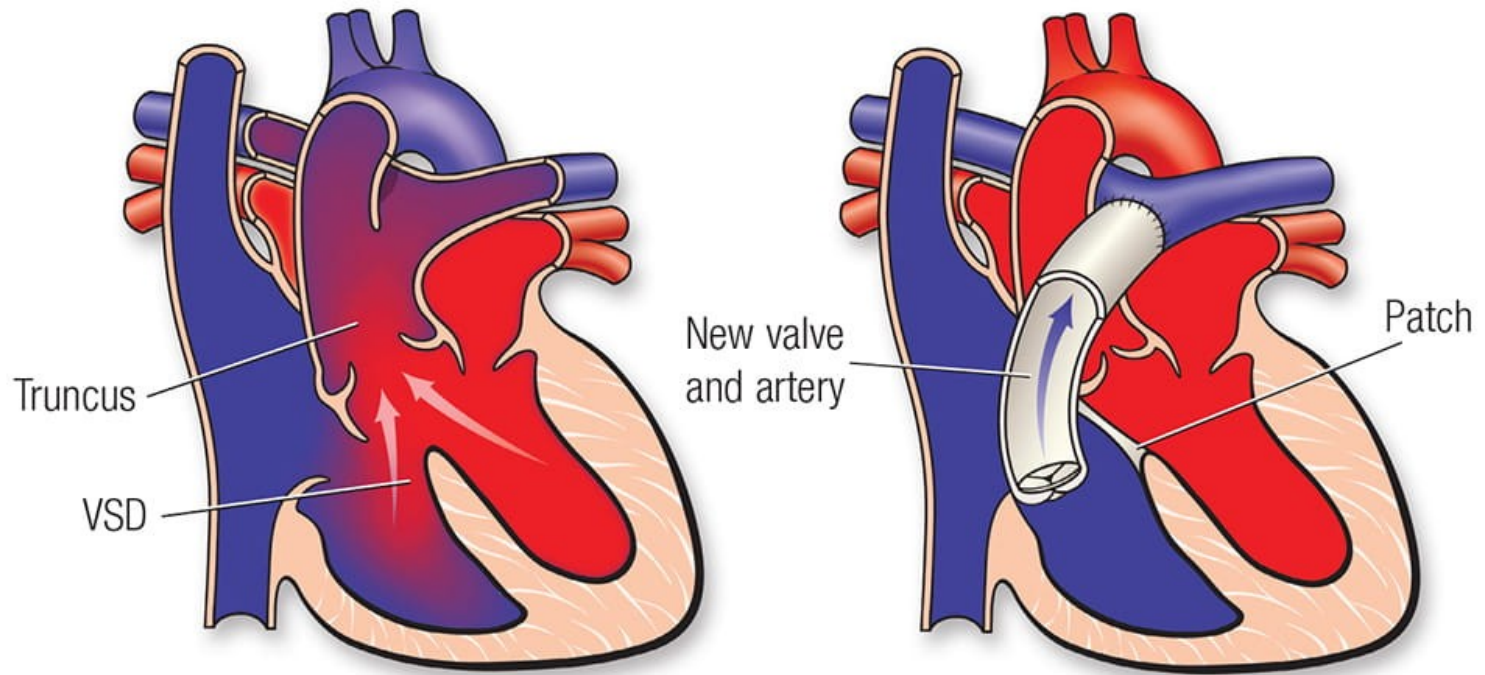
TREATMENT:

- ✓ TGA is a true medical emergency as ductus closes severe hypoxia acidosis and death.
 - PE1 administration immediately on diagnosis. Apnea is the feared complication.
 - Surgery: arterial switch operation (ASO).
- ✓ TGA with VSD predominantly present with sign of congestive heart failure. These patients usually develop pulmonary vascular disease within 3 months so the treatment is anticongestive followed by ASO with closure of VSD before 3 month of age.

3.3. TRUNCUS ARTERIOSUS

- ✓ A single great artery that arises from the base of the heart giving rise to the pulmonary, coronary and systemic arteries.
- ✓ A large VSD is invariably present
- ✓ Total pulmonary blood flow is excessive and effective thus cyanosis is mild..
- ✓ High pulmonary blood flow cause pulmonary hypertension.
- ✓ Volume over load over the left ventricle
→ severe heart failure.

**Rastelli repair with patched septum
and new pulmonary valve/artery**



TRUNCUS ARTERIOSUS

- ✓ Arterial pulse pressure is wide and arterial pulsations are bounding
- ✓ Precordium is hyper active due to the pressure over loaded on RV and volume overloaded LV
- ✓ Harsh and blowing type systolic murmur
- ✓ Increased flow across the mitral valve → apical mid-diastolic murmur.

TRUNCUS ARTERIOSUS...

- ✓ In older children with increased PVR, progressive cyanosis, polycythemia and clubbing may occur.
- ✓ ECG: biventricular hypertrophy
- ✓ CXR: cardiomegaly and increased pulmonary blood flow. Right aortic arch(in 40%), absent thymus→ deGorge syndrome
- ✓ Echo →anatomy well seen with no demonstrable infundibulum.
- ✓ Cardiac catheterization: pass in to pulmonary artery from the truncus. To see intracardiac defect and shunts.

TRUNKUS ARTERIOSUS...

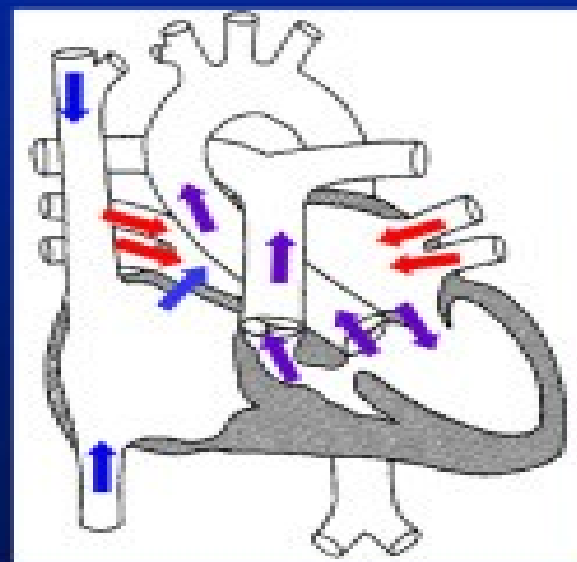
Treatment and prognosis.

- Affected children should be evaluated for hypocalcaemia and T-cell deficiencies associated with DeGorge syndrome.
- Anticongestive treatment but not effective for prolongation of life.
- Surgery: Modified Rastelli procedure before the age of 2-3months.

3.4. TRICUSPID ATRESIA

Tricuspid Atresia

- Complete mixing
- All systemic venous flow must cross atrial septal defect
- All pulmonary blood flow must cross VSD or go through patent ductus arteriosus
- May need a systemic to pulmonary artery shunt
- Eventually will need single ventricle palliation



3.4. TRICUSPID ATRESIA

- ✓ ECG: LVH tall p & R waves.
- ✓ After 2-3 years of age → inverted T Wave.
- ✓ CXR: with small VSD and PS the lung fields are oligemic and heart size is normal.
- ✓ Cardiomegaly and plethoric lung when pulmonary blood flow is un restricted
- ✓ Echo: describes the anatomy.
- ✓ Catheterization :only indicated before the surgery

Treatment and prognosis

- PE1 in new born with markedly decreased pulmonary blood flow to keep the ductus patent.
- Usually indicated in patients with small VSD and partial O₂ saturation (so₂) < 80 mmHg.

✓ Surgery: Fontan procedure

3.5. TOTAL ANOMALUS PULMONARY VENOUS CONNECTION. (TAPVC)

- ✓ Failure of incorporation of the common pulmonary vein into the posterior wall of the LA.
- ✓ Communications with the heart through anomalous connection to SVC which may drain to the coronary sinus or to right atria or directly to the right atria.
- ✓ The clinical presentation depends on the type of anomalous drainage and the presence or absence of pulmonary venous obstruction.

TAPVC

- ✓ Infants with obstructed pulmonary veins exhibit the condition in the newborn period with severe cyanosis and tachypnea in the first few hours of life.
- ✓ For children with mild or no obstruction cyanosis is subtle but may increase with crying or exercise. Heart failure may be very severe, with a Hyper active right ventricular impulse and a widely split 2nd heart sound.

TAPVC

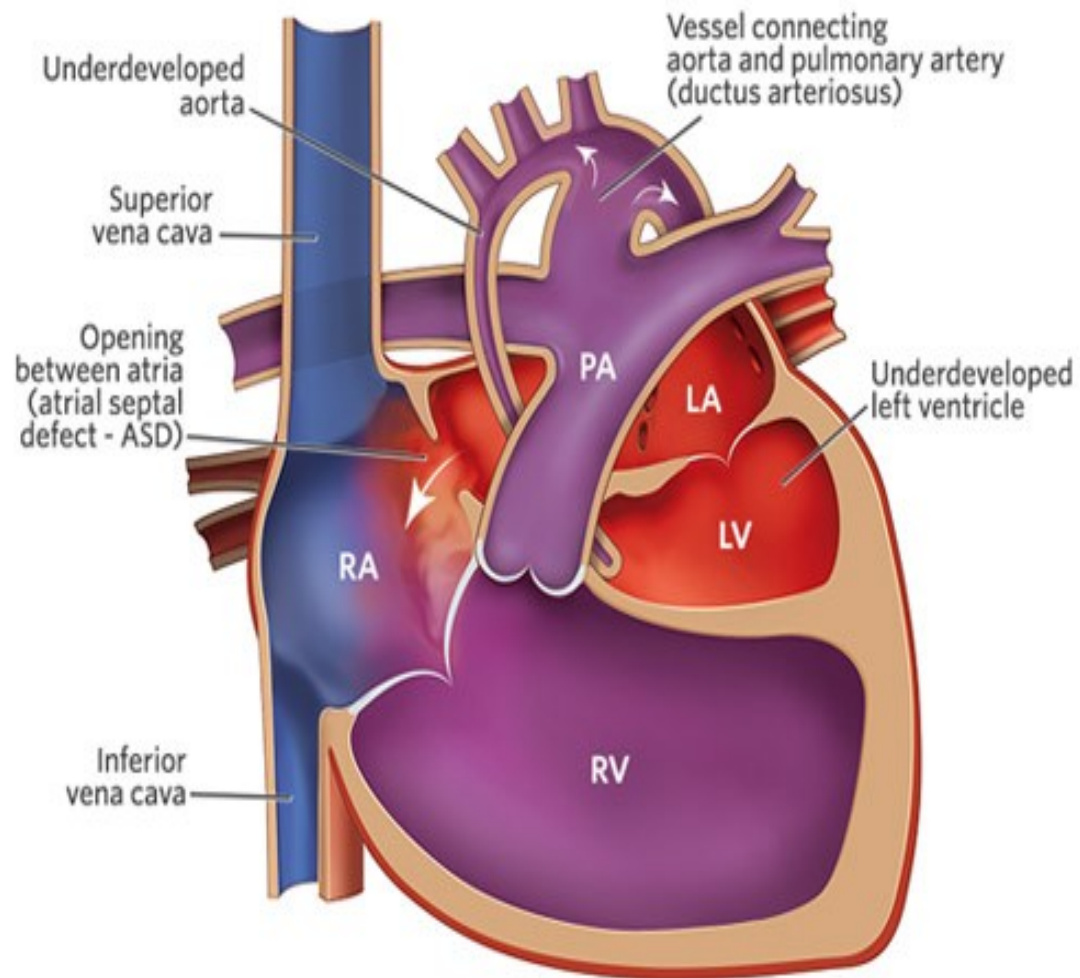
- ECG: marked RVH.
- CXR: marked pulmonary venous congestion with normal heart at the new born which is similar to respiratory distress syndrome.
- Cardiomegaly and increased pulmonary vascularity are seen on older children.
- ECHO. sufficient for the decision of surgical procedure.
- Cardiac catheterization is contraindicated
- Contrast study further exacerbates pulmonary edema & RD

✓ **Treatment:**

- ✓ SURGERY, Opening a wide connection between the common pulmonary vein and the posterior wall of the left atria and ligation of the previous anomalous connections.
- ✓ The prognosis is excellent with a risk of 2% and low incidence of postoperative complication.

3.6. HYPOPLASTIC LEFT HEART SYNDROM (HLHS)

- ✓ Includes a number of closely related anomalies that are part of hypoplasia and the under development of the LV.
- ✓ Mitral and aortic valves usually are hypoplastic or atretic, and ascending aorta is usually small.
- ✓ After birth the pulmonary venous return comes back to the left atria and must go across the atrial septum.
- ✓ With small ASD Pulmonary venous pressure increase and sign of left side heart failure.



3.6.HLHS

- ✓ Constriction of the ductus arteriosus lead to low COP and acidosis.
- ✓ Usually asymptomatic before the ductus closes .Later the child exhibits profound cardio vascular collapse.
- ✓ PE1 should be considered in any infant in shock, particularly if the child has hepatomegaly or poor response to fluid resuscitation.
- ✓ ECG: RVH
- ✓ CXR: cardiomegaly with increased pulmonary vascularity and congestion.
- ✓ ECHO. Diagnostic. Right side of the heart is large.

3.6.HLHS

Treatment

- ✓ Staged operation (3 stages). The overall survival rate is 50%.
- ✓ Because of long term commitment and poor outcome some families choose to refuse any surgery.
- ✓ Cardiac transplantation is an alternative approach.

References

- Nelson Text Book of Pediatrics 21 edition.
- Nelson Essentials of Pediatrics.
- UPTODATE 22